

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	18 July 2026
Team ID	
Project Name	Nutrition Assistant
Maximum Marks	4 Marks

Functional Requirements

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration & Authentication	Registration via form Login with JWT access + refresh tokens Forgot/reset password flow Logout & refresh-token rotation
FR-2	Profile Management	Update demographics, goal & activity level, diet preference Medical conditions & food allergies list Profile photo upload (Cloudinary / local fallback)
FR-3	Calculators	BMI calculator (metric/imperial) with category guidelines Calorie calculator (BMR via Harris-Benedict + TDEE via activity multiplier) Apply calculated calorie goal to profile
FR-4	Food & Water Tracking	Log food per meal slot with macros Auto-recalculate daily nutrition totals Log & delete water intake entries View tracker history by date
FR-5	Recipes & Meal Planning	Search / filter / paginate recipes Create custom recipe with image upload Favorite / unfavorite recipes Plan meals in the Meal Planner
FR-6	AI Nutrition Chatbot	Ask nutrition questions in natural language Receive Gemini-generated or rule-based fallback response Maintain conversation history context
FR-7	Admin Panel	View platform statistics (users, recipes, foods, active logs) Manage users with cascade delete of associated data Create / update / delete standard system foods

Non-functional Requirements

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Responsive, glassmorphic interface built with React 19, Vite and Tailwind CSS v4, including a dark mode toggle, usable across desktop and mobile.
NFR-2	Security	JWT-based dual-token authentication (access + refresh), bcryptjs password hashing, Helmet security headers, CORS allow-list, express-rate-limit throttling, and role-guard middleware protecting admin routes.
NFR-3	Reliability	Zod request validation combined with Mongoose schema validation, plus centralized Express error-handling middleware, maintain data integrity and consistent API responses.

NFR No.	Non-Functional Requirement	Description
NFR-4	Performance	Vite-based fast frontend builds, an Axios API layer with automatic token-refresh interceptor, and indexed MongoDB text search on foods/recipes keep search and dashboard interactions responsive.
NFR-5	Availability	The React client is deployed as a static site while the Express API runs as a persistent Node.js service, backed by a managed MongoDB Atlas cluster.
NFR-6	Scalability	A modular MVC-layered backend (routes → middleware → controllers → models) with role-based access control allows new modules (premium plans, wearable sync, mobile app) to be added without reworking the core platform.